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Electron microscopy of iron oxides and implications for the origin of magnetizations and rock magnetic properties of Banded Series rocks of the Stillwater Complex, Montana

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Abstract. The origins of multiple magnetizations of the Archean Stillwater Complex have been investigated through scanning electron microscopy and scanning transmission electron microscopy observations of mineralogical relations, using representative samples from nine sites in mafic Banded series rocks. On the basis of directional grouping and demagnetization behavior, three magnetizations (here labeled A, B, and C) have been recognized. The natural remanent magnetizaton (NRM) is typically dominated by only one of these magnetizations and multicomponent behavior in individual specimens is rare. The A remanence resides in magnetic grains of high median destructive fields and high, discrete laboratory unblocking temperatures and is inferred to be a primary thermoremanent magnetization, of circa 2.71 Ga age. The B and C magnetizations, of lower median destructive fields and more distributed unblocking temperatures, are inferred to be secondary and related to alteration, including serpentinization, involving limited, moderate to low-temperature fluid interaction, perhaps in response to thermotectonic events (e.g., mafic dike emplacement). Samples with NRM dominated by the A magnetization contain titanium-free magnetite needles (width < 1 μm and maximum dimension between about 2 and 50 μm) with preferred orientations in cumulus plagioclase crystals; these oxides are interpreted to have formed during initial crystallization of cumulus phases. In samples containing only B or C magnetizations, such needles are absent in cumulus plagioclase, whereas secondary Fe-bearing silicates, particularly zoisite and chlorite, are conspicuous. The samples dominated by the B or C magnetizations contain anhedral low-chromian magnetite ($FeCr_{2-x}Fe_xO_4$, $1.5 < x < 2.0$) along fractures in both cumulus plagioclase and pyroxene and in zones of more pervasive alteration of pyroxenes. The paragenesis of these oxides is consistent with an origin related to hydrothermal alteration. Hysteresis data show that plagioclase crystals containing magnetite needles have higher remanent coercivities (H_{cr} = 60 to 100 mT), whereas plagioclase crystals with anhedral chromian magnetite have remanent coercivities less than 60 mT, consistent with their alternating field response. Because oxygen isotope data show no indication of appreciable alteration of host plagioclase, we infer that the absence of magnetite needles and their associated A magnetization are caused by selective alteration of the magnetite needles to nonmagnetic phases in the remagnetized gabbros. In this case, rock magnetic and paleomagnetic data appear to be higher resolution indicators of fluid/rock interaction than the stable isotope data. Anhedral chromian magnetite precipitated during hydrothermal alteration and therefore carries secondary magnetizations.

Introduction

The origin of geologically stable magnetizations in rocks has long been the focus of considerable paleomagnetic and rock magnetic research. Numerous examples in conventional paleomagnetic research demonstrate how the ages of geologically stable magnetizations have been determined relative to processes affecting the rocks (e.g., folding, dike intrusion, erosion). Meanwhile, rock magnetic studies have provided workers with general characteristics of assemblages of magnetized grains that have the greatest probability of carrying geologically stable magnetizations. Major advances in understanding the origin of these magnetizations have occurred when the setting and history of the actual grains inferred to carry the magnetizations have been determined using techniques appropriate for the size of the grains [e.g., *Evans and McElhinny*, 1969; *Morgan and Smith*, 1981; *Geissman et al.*, 1988]. These circumstances provide the best

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opportunity to equate a geologically stable and interpretable magnetization with a specific geologic process. In this contribution, we offer an interpretation of the origin of magnetizations characteristic of Banded Series rocks of the Stillwater Complex, southern Montana, based on a combined electron microscopic, rock magnetic, paleomagnetic, and stable isotopic investigation of selected mafic samples from the Banded Series.

Geology

The Stillwater Complex is well exposed along the northern margin of the Beartooth Mountains, southern Montana. The complex is a mafic-ultramafic layered intrusion with a crystallization age of circa 2.70 to 2.71 Gyr [*Premo et al.*, 1990; *Lambert et al.*, 1985] and is exposed along strike for about 50 km. It is in intrusive contact with Archean metasedimentary rocks on the south and is unconformably overlain or in fault contact with Paleozoic and Mesozoic sedimentary rocks on the north (Figure 1). The complex is divided into three general sequences: the Basal Series, the Ultramafic Series, and the Banded Series. The igneous stratigraphy and petrology of the complex have been intensely studied over many decades. Several stratigraphic subdivisions have been proposed for the complex; here we employ the stratigraphy (Figure 2) introduced by *McCallum et al.* [1980] and *Raedeke and McCallum* [1984] and summarized by *Zientek et al.* [1985]. Alteration of Stillwater Complex rocks, principally as serpentinization of cumulus Fe-Mg silicates [*Page*, 1976], is varied, with the greatest degree restricted to rocks of the Ultramafic Series and olivine-bearing cumulates (troctolites) of the Banded Series. In the context of this contribution, considerable paleomagnetic work has been done on the .complex and related rocks [*Bergh*, 1968, 1970; *Geissman and Mogk*, 1986; J.W. Geissman and C.J. Hale, manuscript in preparation, 1996; J.W. Geissman, unpublished data, 1996]. Banded Series rocks, as discussed in greater detail below, have yielded the most consistent and interpretable paleomagnetic data from the complex and are the focus of the current investigation.

Sampling and Methods

From an extensive paleomagnetic collection, a total of 10 representative cores from 9 sites were studied using scanning electron microscopy (SEM), scanning transmission electron microscopy (STEM), rock magnetic techniques, and stable isotope geochemistry. All samples examined were mafic rocks from the lower and middle Banded Series of the complex (Figures 1 and 2). Thin sections for mineralogical or petrologic study were cut directly from standard paleomagnetic core samples which had been characterized paleomagnetically, revealing geologically interpretable stable primary or secondary magnetizations. As discussed in greater detail below, in the context of this study, we have identified three discrete magnetizations in Banded Series rocks of the Stillwater Complex. The A magnetization is interpreted to have been acquired early in the subsolidus crystallization history of the complex and is thus thought to be a primary, latest Archean magnetization. The other magnetizations, B and C, are, on the basis of paleomagnetic data, interpreted to be secondary and of Proterozoic age. To attempt to limit the possible origins of identified magnetizations, most samples were collected from areas inferred to be free of younger intrusions such as dikes and sills, except for the sample from site B42 which was within 10 cm of the contact with a Proterozoic dike interpreted to carry a B magnetization.

Most samples for paleomagnetic measurements were collected using a portable field drill and oriented using standard methods. Measurements of the natural remanent

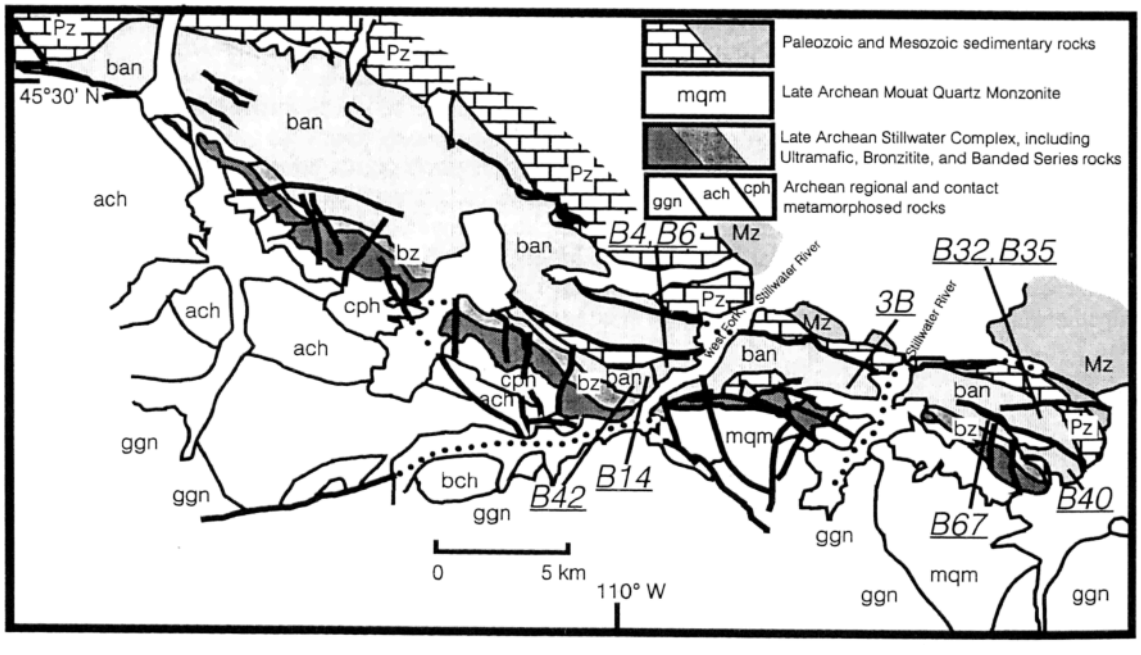


Figure 1. Generalized geologic map of the Stillwater Complex, southern Montana, showing locations of samples form Banded Series rocks examined in this study. Geologic map modified from *McCallum et al.* [1980].

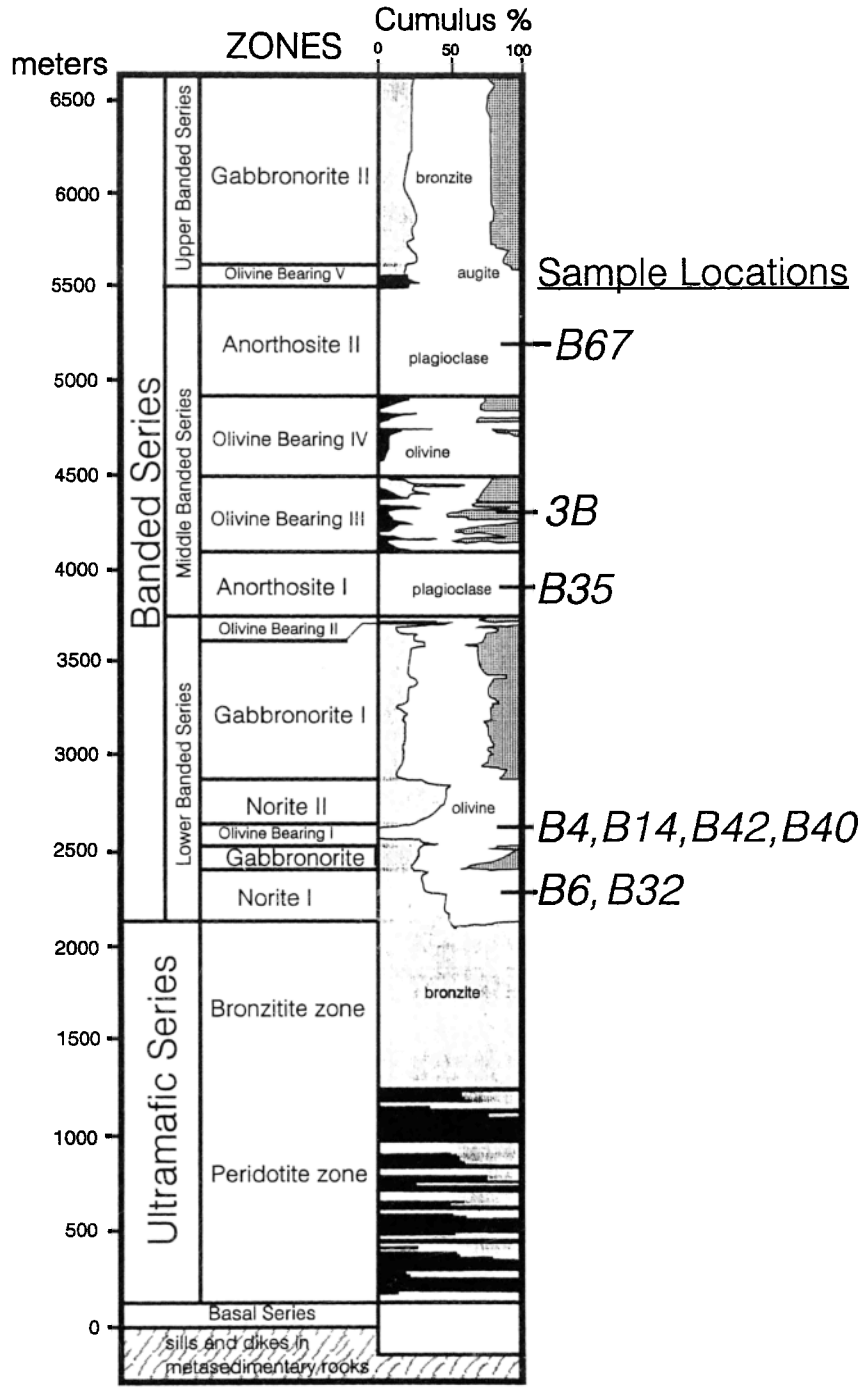


Figure 2. Generalized stratigraphic column for the exposed part of the Stillwater Complex, showing stratigraphic location of Banded Series rocks examined in this study. Column modified from *McCallum et al.* [1980].

magnetization (NRM) employed either a 2G Enterprises superconducting rock magnetometer or Schonstedt spinner magnetometer, both of which are computer interfaced. Alternating field (AF) demagnetization employed either Schonstedt (peak field of 100 mT) or 2G (peak field of 140 mT) instrumentation. Thermal demagnetization employed a Schonstedt unit. Specifically, this contribution reports demagnetization results from either duplicate specimens from the same samples analyzed with SEM and STEM methods or samples adjacent to those analyzed, as well as site mean directions from all pertinent sites. All demagnetization results were inspected using orthogonal demagnetization diagrams and equal area projections. Directions and associated quality statistics of individual magnetizations identified as linear segments (usually stable endpoints) on orthogonal demagnetization diagrams were calculated following *Kirschvink* [1980].

Anhysteretic remanent magnetization (ARM) and isothermal remanent magnetization (IRM) acquisition and AF demagnetization of ARM, NRM, and saturation IRM were

carried out in order to characterize the bulk rock magnetic properties of the samples. Rock magnetic measurements were made on a computer-interfaced Schonstedt SSM-1a spinner magnetometer. ARM was acquired in a 95 mT peak field with a 0.1 mT biasing field. Hysteresis properties of plagioclase crystals containing different kinds of magnetite minerals were measured on an alternating gradient force magnetometer in the Institute for Rock Magnetism at the University of Minnesota.

Both conventional polished thin sections and "sticky-wax" mounted thin sections were made for each core sample and used for petrographic studies by optical microscopy, SEM, and STEM. A Hitachi S-570 SEM with a backscattered electron detector, and a Kevex Quantum detector (energy dispersive spectrometer, EDS, capable of analyzing elements of atomic number > 5) was used. Specimens for transmission electron microscopy (TEM) and analytical electron microscopy (AEM) analyses were prepared by ion-mill thinning. A Philips CM12 STEM equipped with a Kevex Quantum detector and operated at 120 kV was used for the TEM/AEM analyses. The selected area electron diffraction (SAED) patterns and bright-field images were obtained in TEM mode. The AEM analyses were obtained in scanning mode, which gave rise to an electron beam approximately 50 Å in diameter, by rastering the beam (200-1000 Å wide square areas) over thin edges of the specimens. Therefore a grain as small as 200 Å in diameter can be analyzed without significant beam overlap on other grains or other phases. Specimens were tilted toward the detector by 20° giving an X ray takeoff angle of 34°. The X ray spectra were acquired at 100-300 counts per second for 200 s (live time).

SEM analysis cannot distinguish between magnetite, hematite, and goethite because those minerals all contain iron and oxygen and conventional energy dispersive spectrum (EDS) analysis does not provide a quantitative iron:oxygen ratio. In addition, SEM images can only imply specific identifications based on textures. However, the STEM not only has higher magnifications but also provides SAED patterns by which the crystal structure and hence the mineral phase can be definitively identified.

Oxygen isotopic data were obtained from plagioclase separates from samples with different magnetizations. The isotopic measurements were carried out in the Stable Isotope Laboratory at the Department of Geological Sciences of the University of Michigan. The plagioclase crystals were ultrasonically cleaned in dilute HCl. Oxygen was extracted from the samples by the laser fluorination technique using BrF_5 as reagent [*Sharp*, 1990] and converted to CO_2 by reaction with hot carbon. The CO_2 was analyzed on a Delta - S Finnigan MAT mass spectrometer. The results are reported in the usual δ-notation relative to the SMOW standard [*Craig*, 1961].

Paleomagnetic Results

The paleomagnetism of the Stillwater Complex was first studied by *Bergh* [1968, 1970]. Based on limited yet careful (for the time) demagnetization work, Bergh reported a well-defined, well-grouped magnetization in relatively fresh Banded Series rocks from nine sites at two localities. The magnetization reported was only of one polarity and yielded a mean direction having a declination of about 160° and an inclination of about -40° after correction for restoration of igneous layering to the horizontal. He interpreted this magnetization to have been acquired during the initial crystallization of the complex, the age of which was then poorly known. He also obtained well-grouped data (after correction: declination/inclination of about 16°/+26°) from a diabase dike that cuts Banded Series rocks.

More extensive paleomagnetic sampling, initial results of which are reported by *Geissman and Mogk* [1986], has demonstrated the presence of at least three geologically significant magnetizations, referred to here as A, B, and C, in Banded Series rocks of the Stillwater Complex, based on the demagnetization behavior of the NRM and directions of the magnetizations that were isolated. The B and C magnetizations, furthermore, have also been identified in Proterozoic mafic dikes that intrude the complex. At the site level, for the samples examined in this study, magnetizations are acceptably grouped (Table 1). Examples of both AF and thermal demagnetization behavior (Figures 3 and 4), revealed that each of the magnetizations has unique characteristics.

The A magnetization is of dual polarity, with an in situ direction that is east to northeast (or west to southwest) with a generally moderate to steep (30° to 60°) negative (or positive) inclination (Figure 3). Of the sites yielding this magnetization, the east and negative inclination remanence predominates (Table 1). For the sake of possible comparison with paleomagnetic data of comparable age, after restoration of magmatic layering to the horizontal, using present strike axes [*Geissman and Mogk*, 1986], the east and negative magnetization becomes south directed with a moderate to shallow negative inclination, as originally reported by *Bergh* [1968, 1970]. The A magnetization is characterized by high (> 550°C) and discrete laboratory unblocking temperatures (Figure 3) and high (> 50 mT) median destructive fields (Figure 3) and, presumably, coercivities, suggesting that it is carried by fine single-domain (SD) and pseudo-single-domain (PSD) magnetite.

The B magnetization is considerably more dispersed than the other two magnetizations and usually has an in situ direction with a north (or south) declination and moderate to steep positive (or negative) inclination (J.W. Geissman, unpublished data, 1996). Corrected for the orientation of the magmatic layering in the Stillwater Complex, this component has a north-northeast (or south-southwest) declination and positive (or negative) shallow inclination (Table 1). One of the samples (B42C) revealing the B magnetization (note in this case the magnetization is of north-northwest declination and moderate negative inclination) was obtained from rocks in contact with a Proterozoic mafic dike. In comparison to the A magnetization, this magnetization usually has more distributed laboratory unblocking temperatures and median destructive fields generally less than 30 mT.

The C magnetization is also of dual polarity; the predominant in situ direction is usually of southeast declination and shallow inclination (Figure 4). Upon structural correction for magmatic layering this magnetization is of southeast declination and moderate positive inclination (or antipode) (Table 1). In comparison to the A magnetization, this component also has either more distributed laboratory unblocking temperatures or it unblocks over a narrow range of temperatures that are lower than the range typical of the A magnetization. Similar to the B

Table 1. Paleomagnetic Data From Sites in Banded Series Rocks, Stillwater Complex, Used in This Study, as Well as Related Sites

Site	Locality	*n*/*N*/No	In Situ *D*, *I*	Mag.	α_{95}	*k*	Str. Corr.	Corr. *D*,*I*	VGP Lat., Long.
3	Mountain View	9/6/8	57.4,-37.7	A	13.1	16.4	276/92	141.5,-28.2	-45.3,309.8
B4	West Fork	10/8/9	135.3,7.5	C	5.4	82.2	325/45	132.8,-1.5	-28.8,307.8
B6	West Fork	7/5/6	140.8,-18.0	C	13.1	22.1	325/45	155.1,-15.5	-46.3,288.1
B14	West Fork	7/6/7	279.4,59.6	A	7.7	61.8	325/45	10.0,59.9	81.1,13.2
B32	Benbow	7/8/9	61.4,-33.2	A	12.9	22.9	297/95	169.0,-39.8	-65.0,275.6
B35	Benbow	8/7/8	76.0,-27.7	A	6.6	72.1	297/97	155.5,-31.3	-54.5,294.1
B40	Benbow	8/7/8	61.8,-37.7	A	3.8	213.3	297/97	173.7,-34.8	-62.7,264.0
B67	Benbow	8/6/7	57.6,-4.8	A	5.1	117.9	302/100	151.0,-60.5	-68.7,339.2
MA1*	Minneapolis Adit, dike	8/9/9	323.1,55.3	B	7.6	41.9	286/75	348.2,-9.3	38.2,86.0
MA2*	Minneapolis Adit,gabbro	9/9/9	308.8,66.0	B	13.2	12.4	286/75	340.2,1.4	41.5,97.9

The *n*/*N*/No, the number of total specimens (individual magnetization directions) used for statistical analysis from the number of accepted independent samples (*N*) from the total number of independent samples measured (No). *D*, *I*, declination, inclination of the site mean magnetization vector, in degrees east of north and positive downward; Mag., magnetization characteristic of Banded Series rocks at that site; α_{95}, semiangle of the cone of confidence, at the 95% probability level, about the estimated mean direction; the true mean for a population of this dispersion lies within this cone of confidence at the 95% probability level; *k*, best estimate of the dispersion parameter; Str. Corr., orientation of igneous layering measured at that site, with dip direction 90° clockwise from the strike direction (left-hand rule); the structural correction is applied by back-tilt to the horizontal about the present strike axis; VGP is virtual geomagnetic pole position, given as latitude (Lat.) and longitude (Long.) in degrees.

*Site STWB42, where B magnetizations have been identified, represents a contact test between Banded Series rocks and a Precambrian mafic dike, interpreted to be Paleoproterozoic in age. An insufficient number of independent samples with the B magnetization were obtained from the dike and contact rocks at that site, so we provide examples [from *Saxton*, 1989], where the B magnetization was well-identified in a Proterozoic dike and host rock from the Minneapolis Adit.

magnetization, median destructive fields are generally less than 30 mT.

The NRM of over 95% of the Banded Series rocks studied is typically single-component in nature, rendering assessments of the relative age of individual magnetizations difficult if not impossible. In rare cases, either the B or C magnetization is superimposed on the A magnetization, but the opposite behavior has never been observed. In the case of multicomponent behavior, demagnetization experiments reveal that the A magnetization is always carried by an assemblage of grains dominated by relatively high coercivities, whereas either the C or B magnetization resides in grains of lower coercivities.

Oxide Mineralogy: Microscopic Observations

In comparison to the many detailed and systematic studies of silicate mineralogy, relatively little comprehensive work has been carried out on Fe-Ti oxides in Banded Series rocks of the Stillwater Complex. *Maze and Carlson* [1982] reported petrographic observations and electron microprobe data for oxides in Banded Series rocks from the lower Norite and from the upper Banded Zone (Figure 2). Of the distinct types of oxide occurrences described, only one probably has relevance to the paleomagnetism of these rocks because of the overall grain size of the oxides. Maze and Carlson observed 1 to 10 μm low-Ti magnetite and ilmenite grains in what they interpreted to be occasional magmatic glass inclusions in cumulus plagioclase. The composition of the glass inclusions was reported to be similar to cumulus orthopyroxenes in the same samples. The abundance of this type of oxide occurrence appears to be small relative to finer-grained oxides identified in STEM observations, as described below. *Geissman et al.* [1988], in a discussion of paleomagnetic microanalytical techniques, reported limited STEM observations on a single sample of plagioclase from an amphibole/plagioclase pegmatite dike cutting upper Banded Series rocks. Submicron magnetite grains were found in unaltered plagioclase areas. An association between these grains and the remanence isolated from separated plagioclase grains was suggested.

Petrography

All samples studied are coarse-grained mafic rocks mainly composed of cumulus plagioclase and pyroxene. Based on optical microscopy, plagioclase, mainly as a cumulus phase, composes more than 60% (60 to 95%) by volume in all the studied samples (Table 2). Pyroxenes exhibit a range of compositions and textures. Pyroxene minerals can be divided into clinopyroxene and low calcium pyroxene (LCP), including orthopyroxene and pigeonite. Clinopyroxene (cpx) concentration ranges from 5 to 25% (Table 2), and this phase occurs as both cumulus and intercumulus material. Intercumulus clinopyroxene sometimes contains exsolution lamellae of orthopyroxene. LCP concentrations range from 0 to 30% (Table 2). Low calcium pyroxene also occurs as both euhedral cumulus and postcumulus phases. Pyroxene exsolution and inversion features in Banded Series rocks have been described in detail by several workers [e.g., *McCallum et al.*, 1980].

Other interstitial or postcumulus minerals observed in the

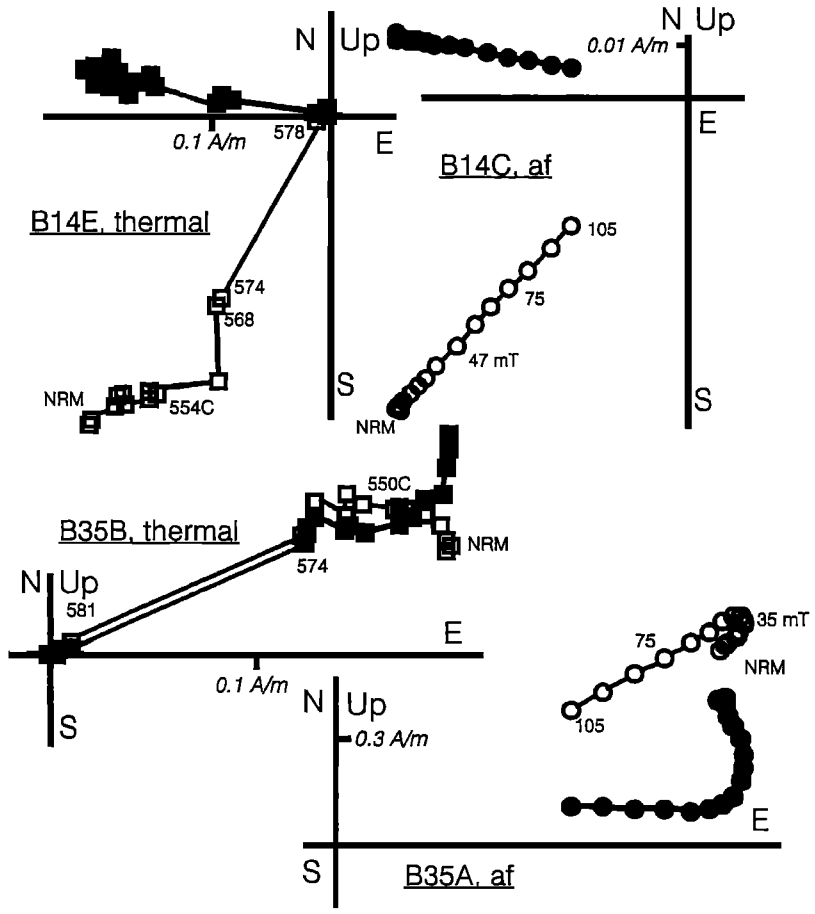


Figure 3. Sets of orthogonal diagrams (in geographic coordinates) illustrating the alternating field and thermal demagnetization behavior of specimens carrying A magnetization, from exact samples used for STEM investigation or specimens from immediately adjacent samples collected at the same paleomagnetic sampling site. For all samples response to both alternating field (circles) and thermal (squares) demagnetization is shown. Projections on the horizontal plane are indicated with solid symbols and those on the vertical plane with open symbols. In all cases, the common axis is the E-W horizontal axis. Peak demagnetizing fields are given in milliteslas and temperatures in degrees Celsius.

samples for this study include ilmenite and sulfides. Ilmenite is found rarely as small grains, typically along grain boundaries between plagioclase and intercumulus pyroxene. Sulfides, tentatively identified as pyrrhotite and chalcopyrite, are found as small patches in interstitial areas.

Optical Microscope and SEM Observations of Iron Oxides

Because conventional EDS analysis does not permit clear distinction among magnetite, hematite, and goethite and because they all have similar fine grain sizes, they cannot be definitely identified using optical microscopy and SEM. We therefore refer to them using the general term "iron oxide."

Four iron oxide grain types/associations have been observed: (1) equant euhedral iron oxide, (2) needle-shaped iron oxide, (3) anhedral fine-grained iron oxide, and (4) mixed iron oxide/iron sulfide assemblages. In addition, ilmenite is found as inclusions in plagioclase and pyroxene grains, as both elongated plates and irregularly shaped plates. Titanomagnetite, as reported by *Maze and Carlson* [1982], was found to be extremely rare in the samples studied. Indeed, only one sample, which does not carry a stable NRM, was found to contain titanomagnetite, implying no obvious correlation between NRM and the presence of titanomagnetite grains.

Euhedral, Ti-free iron oxide was observed in four samples (samples B67D, STW3B, B40I, and B35A) and only within or adjacent to veins in plagioclase crystals (Figures 5a and 5b). The grain sizes range from 5 to 50 μm. EDS analyses indicate that iron is the only detectable cation. The veins also contain Na, K-feldspar, chlorite, zoisite, and amphibole and are abundant in altered plagioclase in these samples.

Needle-shaped iron oxide grains are found in fresh to relatively unaltered plagioclase crystals in samples B67D, STW3, B40I, B14C, B32H, and B35A and generally have a width less than 1 μm and a length between 2 and 50 μm (Figure 5c). EDS analyses show iron as the only cation. Optical observations reveal that most of the needles are elongated either parallel or perpendicular to the {010} albite twin plane of plagioclase crystals (Figure 5d).

Anhedral, fine iron oxide grains are commonly observed on fracture surfaces in plagioclase (Figure 5e), along alteration veins in cumulus plagioclase crystals near areas of LCP

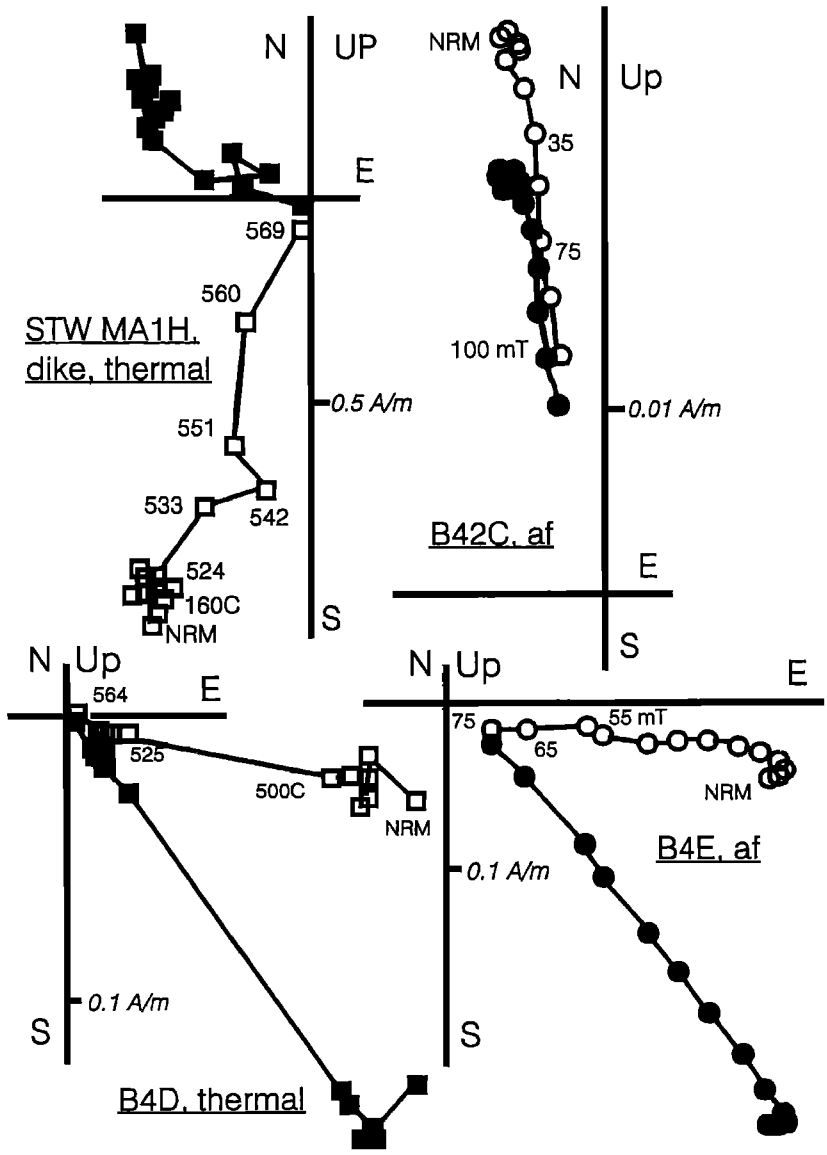


Figure 4. Sets of orthogonal diagrams (in geographic coordinates) illustrating the alternating field and thermal demagnetization behavior of specimens carrying B magnetization (samples B42C and STW MA1H [from *Saxton*, 1989]) and carrying C magnetization (samples B4E and B4D). Conventions as in Figure 3.

alteration (Figure 5f), and within areas of LCP alteration (Figure 5g). The maximum dimension of these grains ranges from 0.01 to 5 μm. EDS analyses indicate that these iron oxide grains are characterized by notable yet variable Cr contents. The texture of the iron oxide grains and their coexistence with areas of cpx and LCP alteration all are interpreted to indicate that these grains formed during hydrothermal alteration of cpx and LCP.

Also, iron oxide grains have been observed in association with pyrrhotite and chalcopyrite in almost all samples studied (Figure 5h). Sulfide minerals are associated with secondary epidote/zoisite replacing plagioclase, chlorite replacing cpx, sphene, albite, and other alteration products. The iron oxide grains associated with sulfides range from 1 to 50 μm in the largest dimension.

Table 2. Estimates of Major Rock-forming Mineral content by Optical Microscope Observations

Samples	pl	cpx	opx	Other Comments
B14C	90	10	-	> 50% cpx has been altered
B35A	95	5	-	> 80% cpx has been altered
B40I	87	10	3	small amount Na-K-feldspar
B32H	65	10	25	small amount Na-K-feldspar
B67D	60	10	30	small amount Na-K-feldspar
STW3B	60	20	20	small amount Na-K-feldspar
B4E	70	20	10	
B4F	60	25	15	
B42C	90	8	2	
B6E	80	12	8	

STEM Observations, Iron Oxide Grains

Fine, needle-shaped iron oxide grains within unaltered plagioclase crystals have been identified as magnetite (Figures 6a, 6b, 6c, and 6d) by SAED patterns using the STEM. AEM reveals that the needles are essentially devoid of titanium. SAED patterns and dark field images (e.g., Figures 6c and 6d) show that the magnetite needles are homogeneous, well-

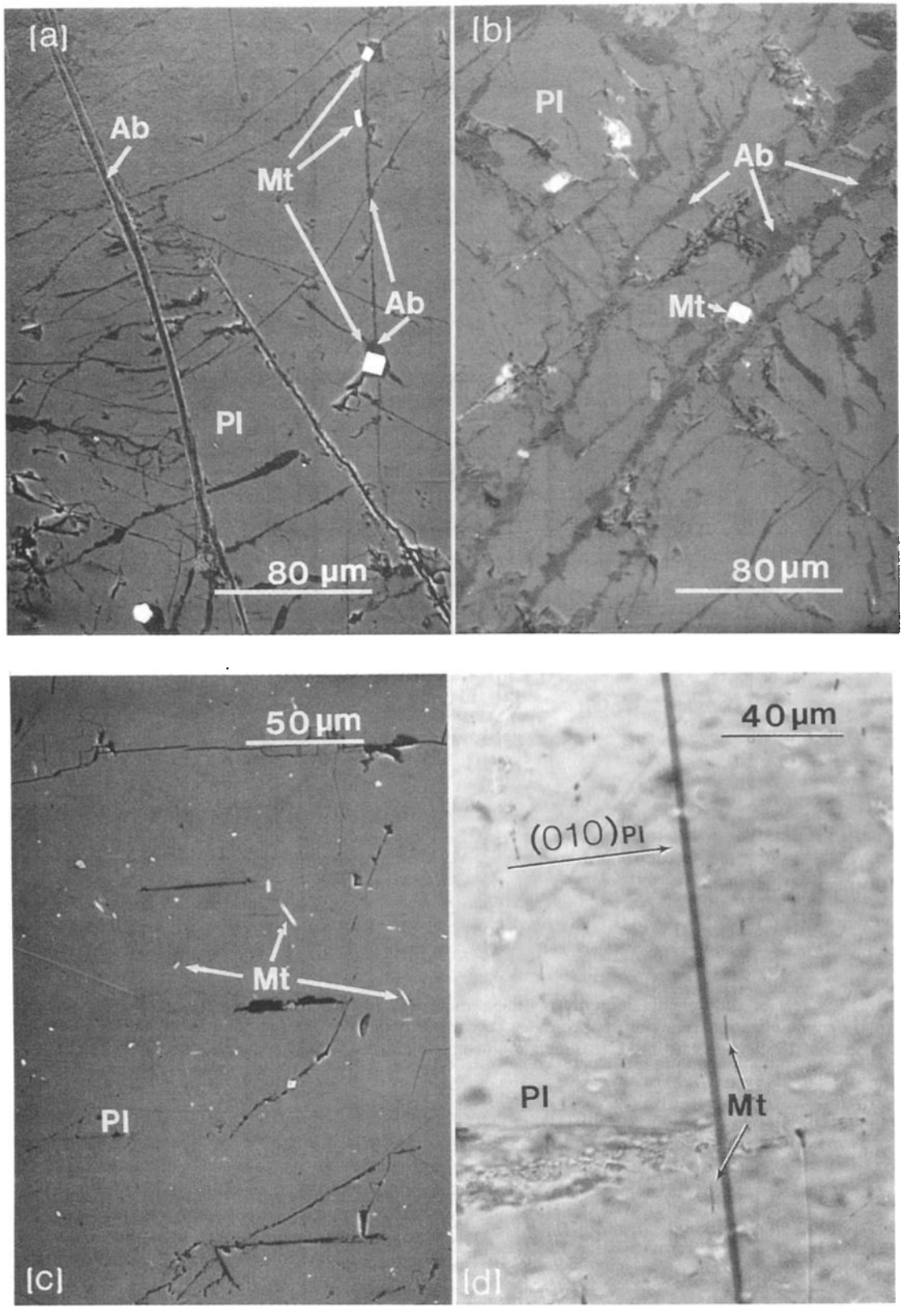


Figure 5. SEM and optical microscope images of representative iron oxide minerals. (a) back-scattered electron (BSE) photograph showing cubic euhedral magnetite in albite veins, sample B67D; (b) BSE photograph showing cubic euhedral magnetite in albite veins, sample STW3B. Mt, magnetite; Ab, albite; pl, plagioclase; (c) BSE photograph showing magnetite needles in plagioclase, sample B14C; (d) optical microscope photograph (transmitted light, crossed nicols) of magnetite needles in plagioclase, showing magnetite needles parallel to (010) twin plane of plagioclase. Mt, magnetite; pl, plagioclase; (e) SEM photograph showing anhedral iron oxide grains on a fracture surface of a plagioclase crystal, 50% BSE + 50% SEI, sample B4F; (f) BSE photograph showing amorphous iron oxide minerals in pore spaces of plagioclase (ion milled sample of B4F); pl, plagioclase; Cr-Mt, chromian magnetite; (g) BSE photograph showing LCP replaced by talc and iron oxides in sample B4F; (h) BSE photograph showing iron oxide minerals (grey) adjacent to iron sulfide minerals (bright), sample B14C). Fe, iron oxide.

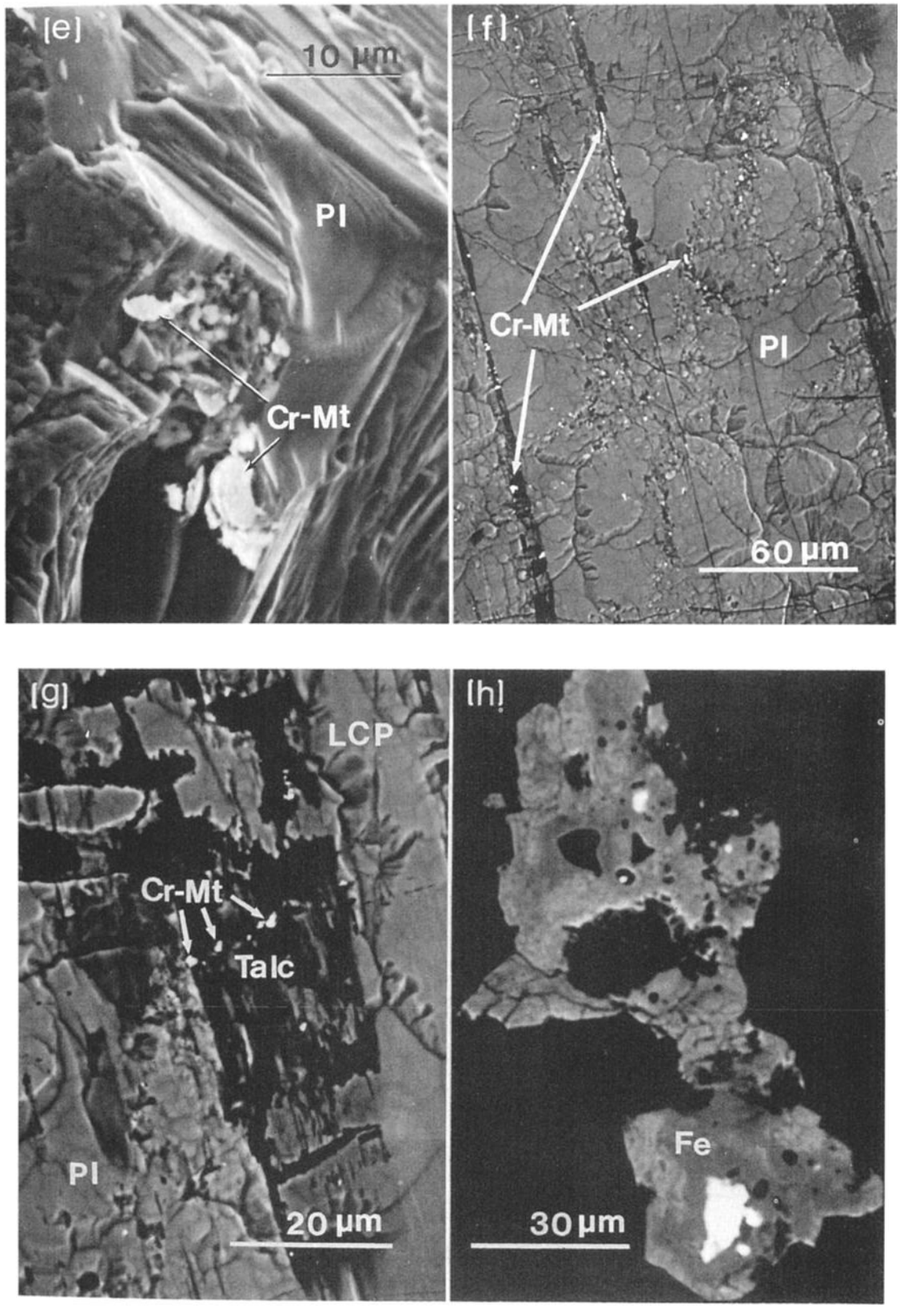


Figure 5. (continued)

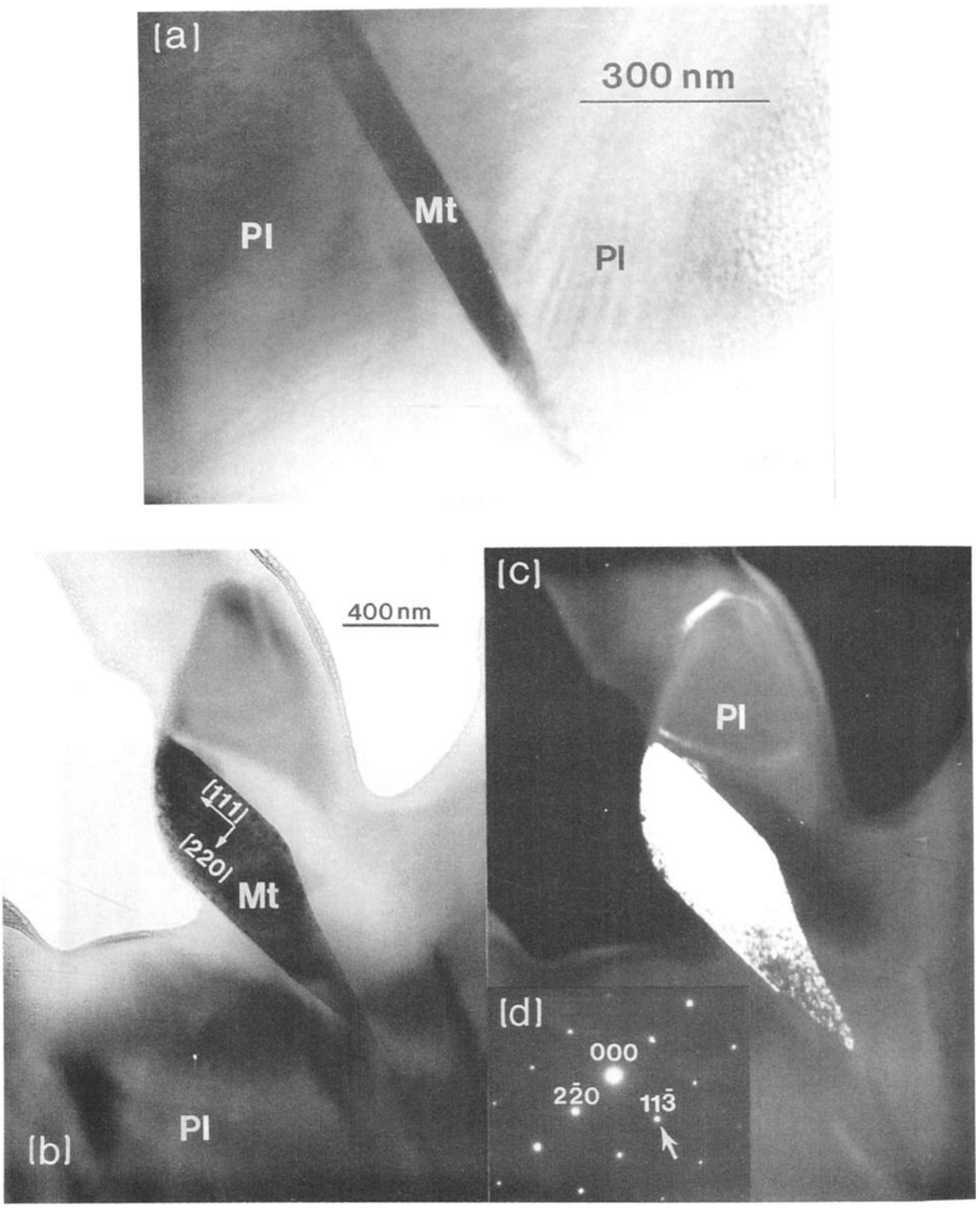


Figure 6. Transmission electron microscopy images of magnetite grains. (a) STEM bright field image of a needle-shaped iron oxide within a plagioclase crystal, sample B35A; pl, plagioclase; Mt, magnetite; (b) STEM bright field image of a needle-shaped iron oxide within a plagioclase crystal, in sample B35A; (c) STEM dark field image of the same magnetite needle as the bright field image of Figure 6b within plagioclase, using (113) reflection; (d) SAED pattern showing that the needle-shaped iron oxide is single-crystal magnetite; pl, plagioclase; Mt, magnetite; (e) STEM bright field image of anhedral chromian iron oxide grains in a secondary amphibole group mineral, sample B4F; (f) STEM bright field image of anhedral chromian iron oxide grains in a secondary amphibole group mineral, sample B4F; (g) SAED pattern for the grain of Figure 6f (arrow pointed) indicating that the anhedral chromian iron oxide mineral is magnetite; Cpx, clinopyroxene; Am, actinolite; Cr-Mt, chromian magnetite; (h) STEM bright field image showing a chromian magnetite grain along a fracture in Cpx, sample B4F. Cpx, clinopyroxene; Cr-Mt, chromian magnetite.

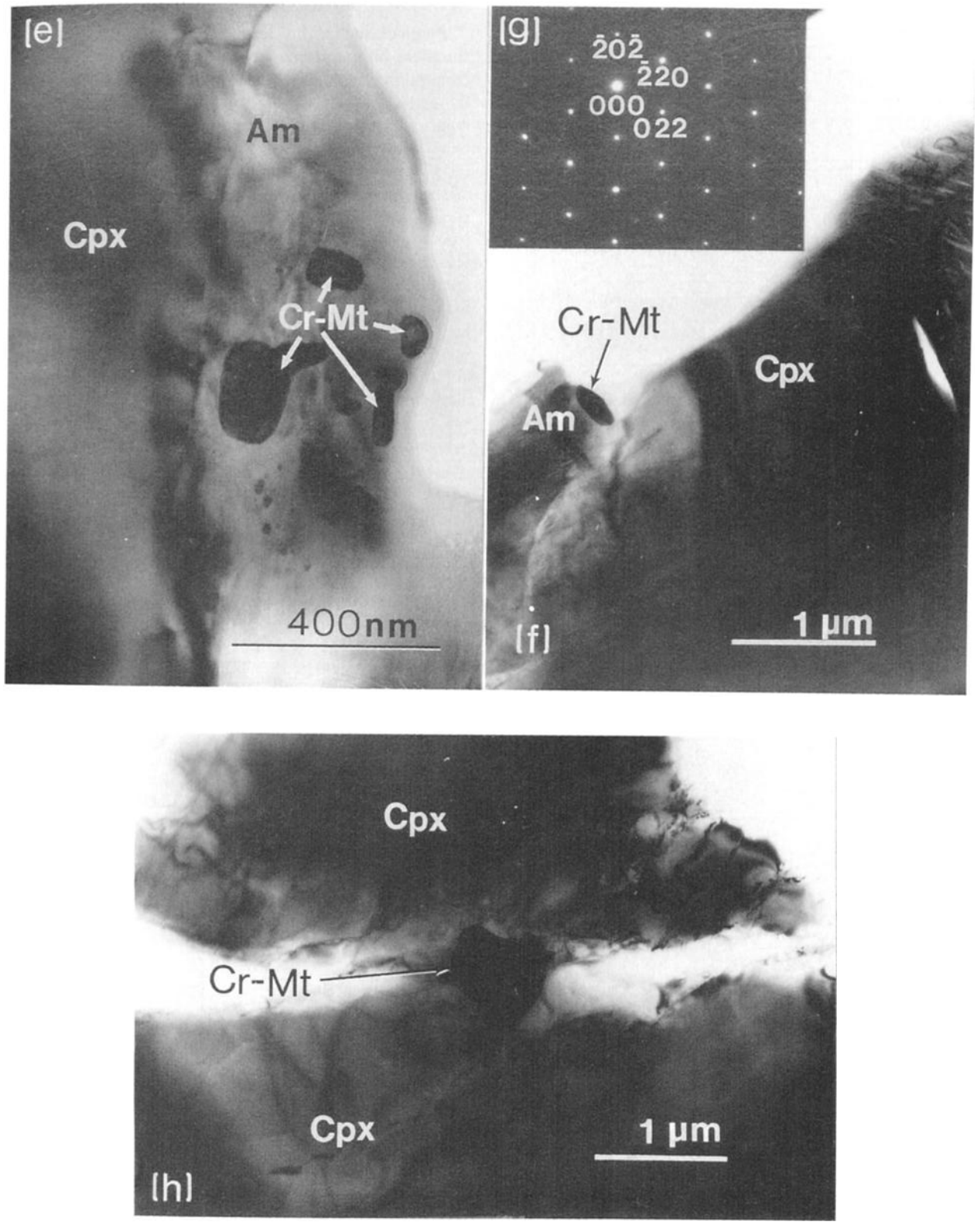


Figure 6. (continued)

crystallized single crystals with [111] as the elongated direction parallel to the (010) plane of plagioclase, consistent with optical microscopic observations (Figure 5d).

Anhedral Cr-bearing iron oxide grains within cpx alteration zones are identified, through SAED patterns, as well-crystallized cubic phases (Figures 6e, 6f, and 6g) and AEM analyses indicate variable Cr contents, generally between 1 and 10% by weight ($FeCr_{2-x}Fe_xO_4$, $1.5 < x < 2.0$) (Table 3). With Cr content less than 10%, this phase, like pure magnetite, has an inverse spinel crystal structure and is magnetic [*Robbins et al.*, 1971] and therefore will be referred to as a low-chromian magnetite. Whenever low-chromian

Table 3. Typical AEM Analyses of Chromian Magnetite in the Stillwater Gabbro

Cation	mt1	mt2	mt3	mt4	mt5	mt6
Mg	0.000	0.000	0.000	0.000	0.072	0.000
Ti	0.015	0.000	0.000	0.000	0.038	0.000
V	0.016	0.033	0.000	0.000	0.000	0.027
Cr	0.321	0.111	0.232	0.195	0.435	0.366
Mn	0.014	0.033	0.054	0.046	0.066	0.090
$Fe^{2+\#}$	3.433	3.718	3.598	3.662	3.134	3.293
Total	3.800	3.895	3.844	3.903	3.745	3.776

Cations are calculated on the basis of four O.
Fe^{2+} represented as total iron.

magnetite grains are observed in association with cpx-amphibole boundaries, they are always within secondary amphibole (Figures 6e and 6f), indicating their secondary origin. Low-chromian magnetite grains occur along fractures and veins in cpx crystals (Figure 6h).

SEM Observation of Alteration of Silicates

Plagioclase is the least altered primary phase in the samples, although the extent of alteration is quite varied. Commonly, fresh and completely altered plagioclase coexist in the same thin section. Fresh plagioclase grains are gray in color; with increasing alteration the grains have a mottled gray and white appearance. Secondary minerals replacing plagioclase include albite and epidote, as determined by EDS analysis (Figures 7a and 7b). This alteration sometimes is spatially related to areas of cpx and/ or LCP alteration. The formation of zoisite/epidote in plagioclase requires iron enrichment relative to fresh plagioclase, possibly from the alteration of cpx and/or LCP or from hydrothermal fluids. LCP is susceptible to hydrothermal alteration and generally has been variably altered. Almost all the LCP crystals observed in the samples have been partially replaced by talc, chlorite, and iron oxides (Figure 7c). In areas adjacent to LCP alteration, anhedral chromian magnetites precipitated in pore spaces of cpx and within cracks of plagioclase. Where cpx is ostensibly replaced by chlorite (Figure 7b), the alteration is typically associated with the replacement of plagioclase by albite and titanium oxide (probably rutile) that is itself

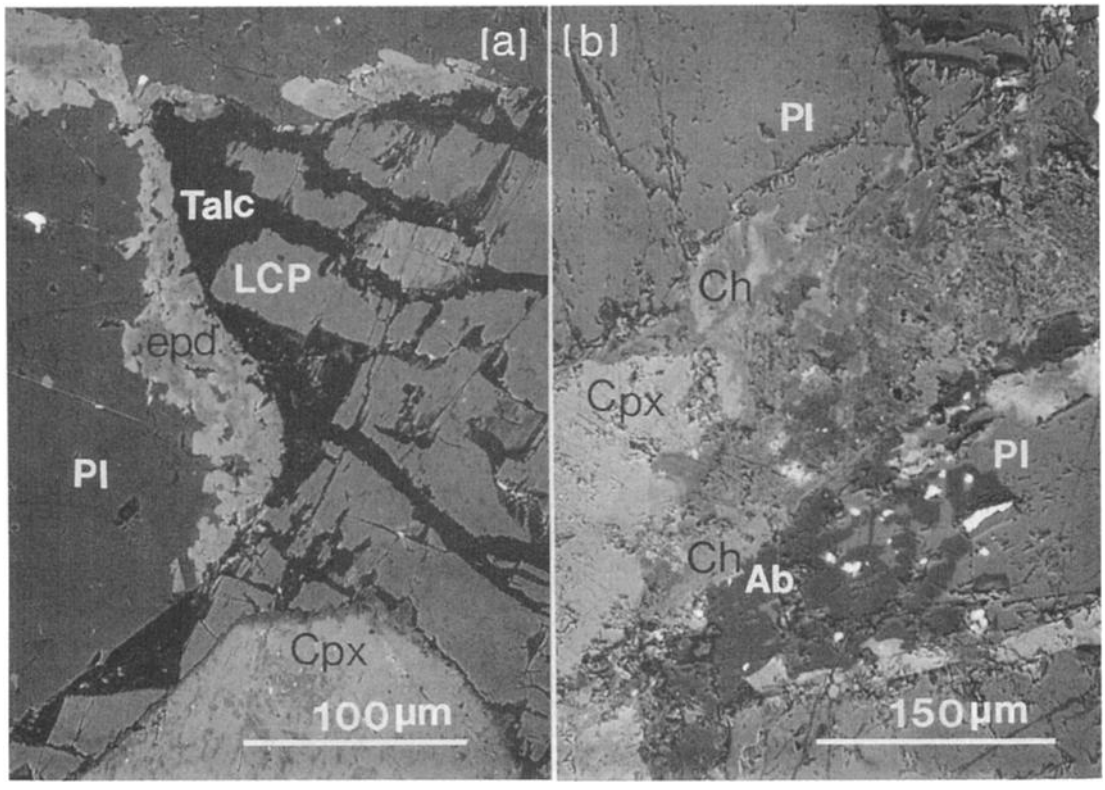


Figure 7. Scanning electron microscopic images of silicate mineral alteration assemblages. (a) BSE photograph showing plagioclase being replaced by an epidote mineral in sample B4E; (b) BSE photograph showing plagioclase replaced by albite and cpx replaced by chlorite in sample B67D; LCP, low calcium pyroxene; pl, plagioclase; Cpx, clinopyroxene; Ch, chlorite; Ab, albite; (c) BSE photograph showing LCP replaced by talc and iron oxides in sample B67D; (d) BSE photograph showing cpx replaced by amphibole and iron oxides in sample B4E. LCP, low calcium pyroxene; Cpx, clinopyroxene; act, actinolite; pl, plagioclase.

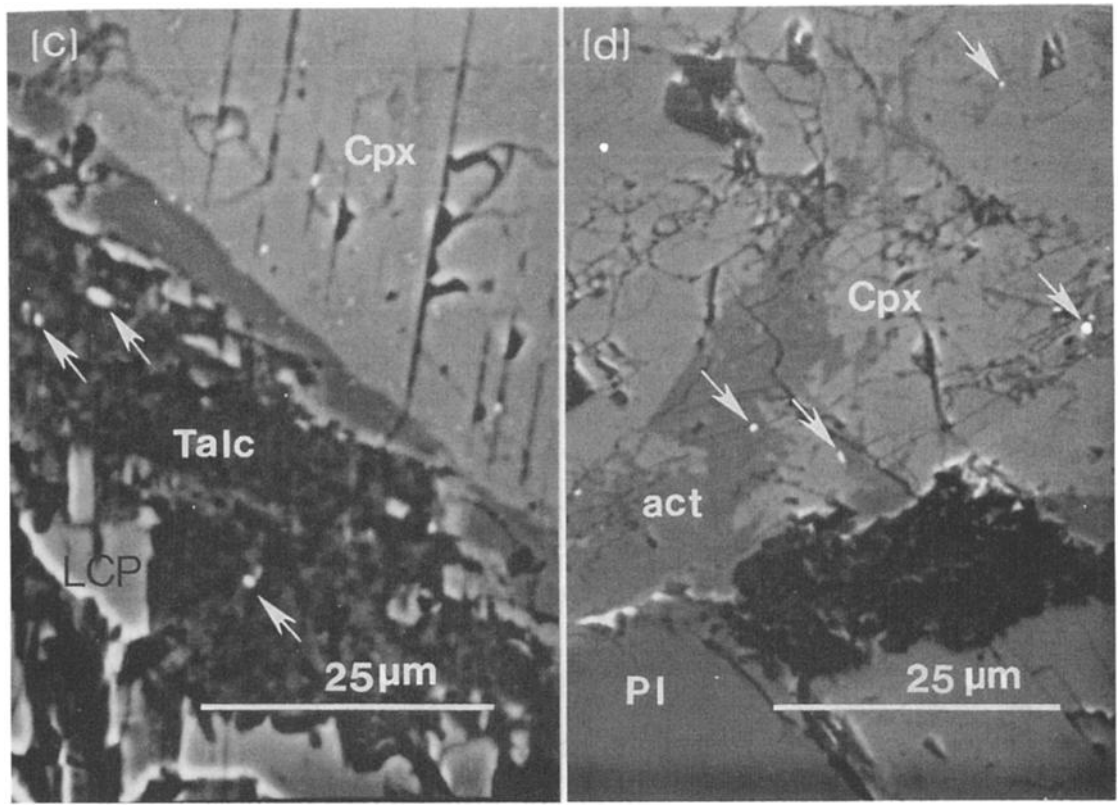


Figure 7. (continued)

replaced by sphene. Partial replacement of cpx by amphibole has been observed in some samples (Figure 7d).

Rock Magnetic Measurements

In IRM acquisition and backfield demagnetization, a cubic magnetic phase is revealed as the principal magnetic phase present in specimens from all samples. All specimens have IRM acquisition curves that are concave downward for steady fields between 0 and about 0.2 T and have coercivity of remanence values less than 150 mT. The slope of the backfield demagnetization curve is constant over a wide range of steady demagnetizing fields, suggesting the absence of two or more populations of magnetized grains of discrete coercivities. Samples carrying A magnetization (Figures 8a and 8b) have higher remanent coercivity (> 100 mT), whereas samples carrying secondary magnetizations have lower remanent coercivity (< 80 mT) (Figures 8c and 8d).

In IRM acquisition, all specimens saturated between 0.2 and 0.35 T (Figure 8). The SIRM acquisition curves indicate that the major magnetic phase in these samples is magnetite, consistent with thermal demagnetization results that show that the maximum unblocking temperature (Figures 3 and 4) is less than the Curie temperature of magnetite (about 580°C). The modified Lowrie-Fuller test [*Lowrie and Fuller*, 1971; *Dunlop et al.*, 1973; *Johnson et al.*, 1975] reveals that ARM is considerably more resistant than SIRM at low to intermediate fields in AF demagnetization (Figure 9), yet somewhat less resistant at very high fields. This behavior has been described as "bimodal" by *Dunlop* [1983], implying two populations of relatively discrete magnetic phases, with the finer-grained, higher-coercivity population dominating the ARM and the coarser-grained, lower-coercivity population swamping the SIRM. Overall, this behavior is consistent regardless of whether the samples contain an A, B, or C magnetization.

Hysteresis properties were measured on rock chips and individual mineral grains studied using an SEM (e.g., plagioclase crystals from samples where Ti-free magnetite needles were identified). The measured values of M_{rs}/M_s for all the plagioclase crystals with magnetites range from 0.05 to 0.41, the ratios of H_{cr}/H_c from 1.5 to 3.5. For hysteresis ratios of M_{rs}/M_s versus H_{cr}/H_c, all values fall in the pseudo-single domain region (Figure 10a). A clear correlation between hysteresis parameters and NRM characteristics exists for most samples. Although there is only slight overlap between samples carrying A magnetization and samples carrying B and C magnetizations (Figure 10a), plagioclase crystals from samples carrying A magnetization alone or in combination with another superimposed magnetization yield distinctly higher coercivity values than plagioclases from samples carrying B and C magnetizations (Figure 10b). Plagioclase crystals with magnetite needles from the samples dominated by the A magnetization have H_{cr} values higher than 60 mT and M_{rs}/M_s ratios generally above 0.2. A few specimens from samples B67D and B35A have H_{cr} values

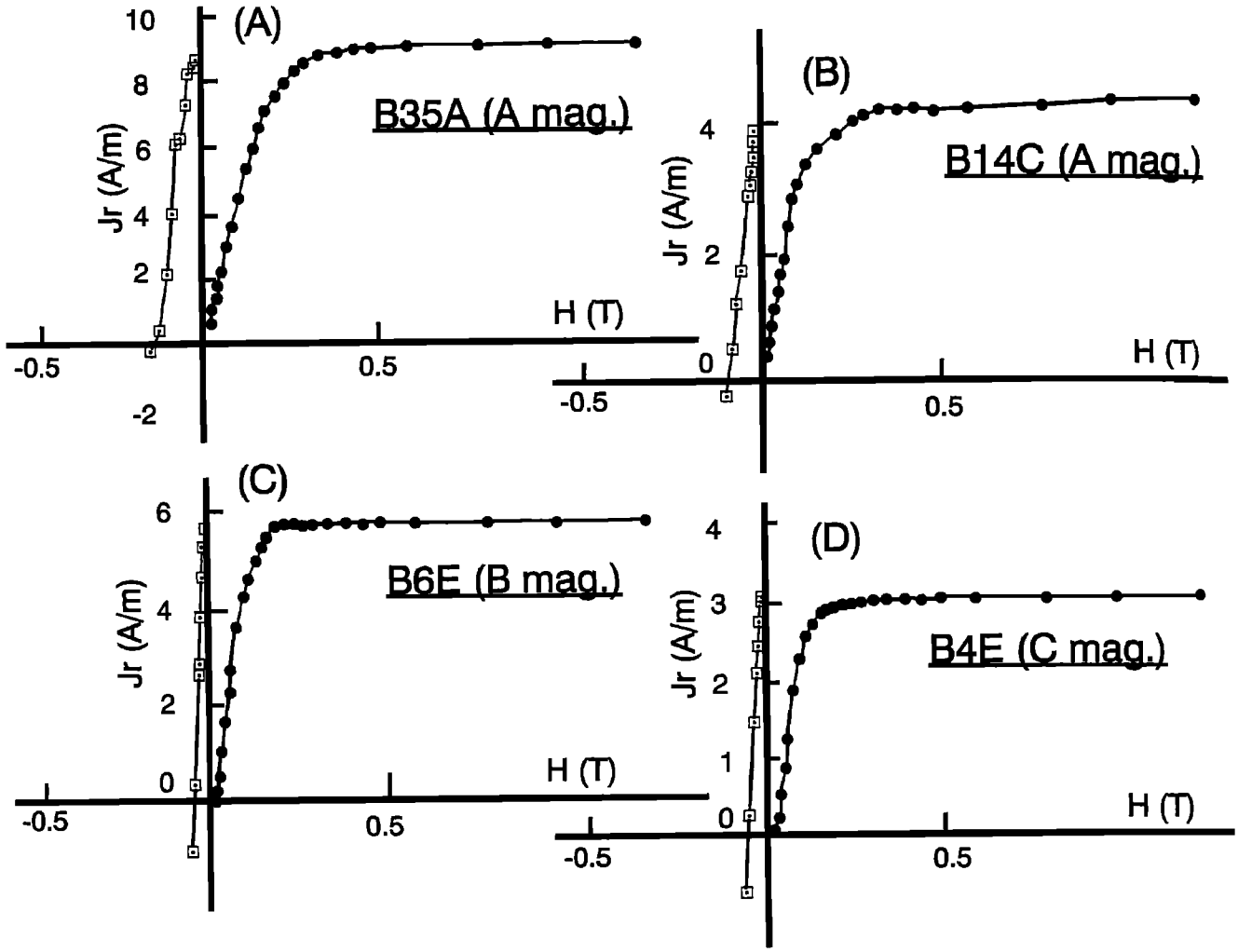


Figure 8. SIRM acquisition curves and backfield AF demagnetization curves of Banded Series samples. (a) and (b) Samples carrying primary A magnetization and (c) and (d) samples carrying secondary C magnetization.

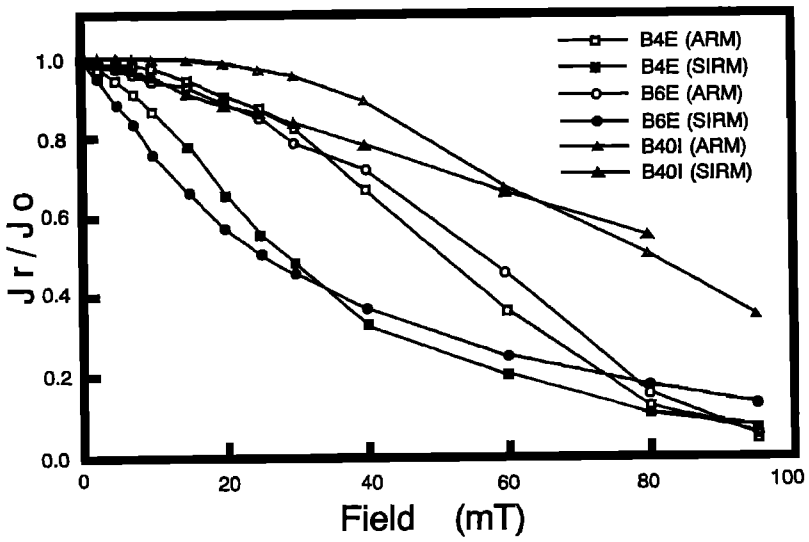


Figure 9. Stability of saturated isothermal remanent magnetization (SIRM; solid symbols) and anhysteretic remanent magnetization (ARM; open symbols) in alternating field (AF) demagnetization (modified Lowrie-Fuller test [*Lowrie and Fuller*, 1971; *Johnson* et al., 1975]), showing SD/PSD behavior in Banded Series samples.

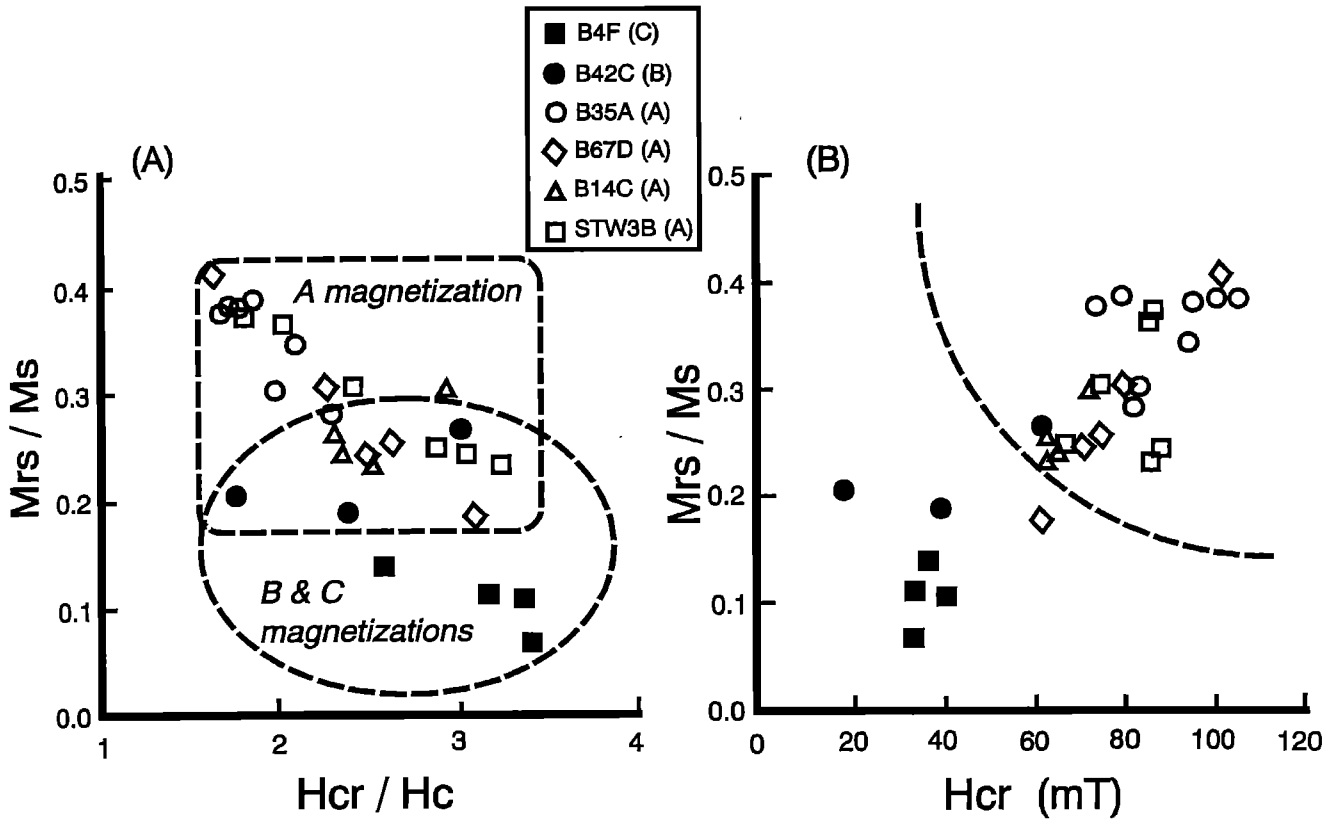


Figure 10. Hysteresis parameters of samples carrying primary magnetization A (open symbols) and samples carrying secondary magnetizations B and C (solid symbols). (a) Plot of M_{rs}/M_s versus H_{cr}/H_c, showing both types of materials contain SD/PSD magnetites [*Day et al.*, 1977; *Dunlop*, 1986]. (b) Plot of H_{cr} versus M_{rs}/M_s, showing distinct differences between samples carrying primary magnetization A and samples carrying secondary magnetizations B and C.

greater than 100 mT and M_{rs}/M_s ratios greater than 0.35 (Figure 10b), indicating that many magnetite needles are in or near the range of single domain magnetite [*Evans and McElhinny*, 1969]. The H_{cr} values for plagioclases from samples without magnetite needles (e.g., samples B4F and B42C, with magnetizations C and B, respectively) are less than 60 mT (Figure 10b), consistent with the backfield demagnetization data from standard specimens (Figure 8). For samples carrying two magnetizations (magnetization A and another poorly defined, geologically insignificant component), the plagioclase grains contain magnetite needles and generally have high H_{cr} values and high M_{rs}/M_s ratios. Sample B67D carries the A magnetization and another unidentified low coercivity component. Most plagioclase crystals from this sample have H_{cr} values greater than 60 mT, consistent with SEM observation that all these crystals contain magnetite needles. One plagioclase crystal from the sample yielded coercivity values of H_c = 15.3 mT, H_{cr} = 40 mT, and an M_{rs}/M_s ratio of 0.209, and this is consistent with SEM observations that this plagioclase contains coarser, euhedral iron oxide grains in veins, as well as fine magnetite needles (Figure 5a), resulting in more intermediate hysteresis properties. Data from this crystal were not included in Figures 10a and 10b. An overall correlation between SEM observations and magnetic measurements indicates that the euhedral iron oxide grains are magnetite of a PSD character and carry the lower coercivity magnetization whereas the magnetite needles in plagioclase crystals are of a SD character and carry the high-coercivity A magnetization.

Stable Isotope Geochemistry

Based on microscopic observations, the conspicuous absence of magnetite needles in cumulus plagioclase in samples with secondary magnetizations suggests that early formed magnetite needles, arguably of early origin, were partially to completely altered to chlorite and epidote, and the majority of the host plagioclase remained essentially unaltered. It is possible that chromian magnetite was precipitated along fractures within the plagioclase during alteration of magnetite needles. To evaluate the extent of fluid/silicate interaction during alteration, oxygen isotopic data were obtained from single plagioclase crystals, with minor concentrations of magnetic phases from samples that carry the different magnetizations.

Because hysteresis measurements reveal the general magnetic characteristics of plagioclase grains and hence the nature of magnetic phases in these grains, samples analyzed in this study were selected from mineral chips on which hysteresis measurements have been made. The results (Table 4 and Figure 11) show a narrow range of $\delta^{18}O$ values for all plagioclase chips, between 5.3 and 6.4‰ (SMOW), which is

Table 4. Oxygen Isotope Results of Plagioclases With Minor Magnetic Minerals as Inclusions

Sample	Color	$\delta^{18}O$ %, SMOW	Magnetic Components
B4E1	white	5.9	C
B4F1	white	5.9	C
B6E1	white	6.4	C
B42C3	white	5.4	B
B14C1	white with grey zones	5.3	A
B14C2	white with grey zones	5.5	A
B14C3	white with grey zones	5.4	A
B34A4	grey	6.0	A+?
B40I1	light grey	6.1	A+?
B67D4	light grey	6.1	A+?
STW3B7	spotty grey	6.0	A+?

+? means a poorly defined component is present in addition to A.

consistent with igneous values expected for and observed on plagioclase from mantle-derived mafic igneous rocks [*Kyser*, 1986] and also consistent with previous work on the complex [*Dunn*, 1986]. *Dunn* [1986] reported 28 plagioclase analyses from Banded Series samples spanning the complex. The values obtained, from 4.8 ± 0.1‰ to 8.0 ± 0.1‰, were interpreted to indicate that the complex had retained its magmatic isotopic composition. Relatively subtle lowering of $\delta^{18}O$ in olivine-bearing zones in the Banded Series was furthermore interpreted to indicate partial interaction with a late stage, high-temperature fluid. Our data show no obvious correlation between the dominant magnetization present and $\delta^{18}O$ values (Figure 11), indicating that the plagioclase crystals with secondary, anhedral chromian magnetite have not substantially reequilibrated with hydrothermal fluids, consistent with microscopic observations. Variations in the modal abundance of iron oxides, of any origin, in these plagioclases are too small to result in systematic variations in the oxygen isotope data.

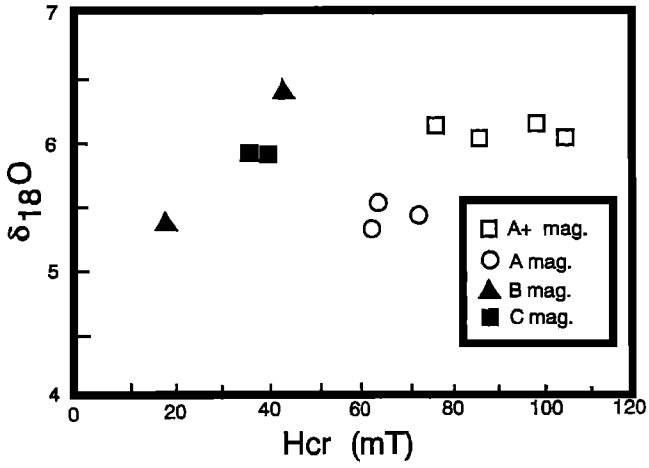


Figure 11. $\delta^{18}O$ (‰ SMOW) versus H_{cr} for plagioclase crystals with trace amount magnetites from samples that carry different magnetizations.

Discussion and Conclusions

Correlation Between Characteristic Magnetic Mineral Assemblages and Paleomagnetism

Based on demagnetization characteristics, the samples examined can be divided into three groups: (1) Samples with only high coercivity and high unblocking temperature A magnetizations (sample: B14C), (2) samples carrying component A as well as a minor low-intensity, superimposed magnetization (samples B40I, B32H, B35A, B67D, and STW3B), and (3) samples carrying only B (sample B42C) or C magnetizations (samples B6E, B4E, and B4F). A comparison of the observed iron oxides and the magnetizations isolated in these samples (Table 5) suggests several correlations between magnetic mineral genesis and remanence properties.

Chromium-bearing magnetite is not a common magnetic phase in geologic materials and relatively little is known of its magnetic properties and their variations with composition. *Robbins et al.* [1971] demonstrated synthetic Cr spinels to be characterized by the molecular formula $FeCr_{2-x}Fe_xO_4$ ($0 \le x \le 2$). The magnetic properties of this system ($FeCr_{2-x}Fe_xO_4$) vary among three compositional ranges: $0 \le x < 0.68$; $0.68 \le x \le 1.38$; and $1.38 \le x \le 2.0$. The compositions of low-chromian magnetite observed in the Stillwater gabbros lie mostly in the third region, where the Curie temperature decreases linearly with increasing Cr^{3+} substitution from about 580°C (pure magnetite) to 380°C for chromian magnetite ($x = 1.38$) [*Robbins et al.*, 1971, Figure 1b]. The magnetic moment also decreases with increasing Cr^{3+} from 4 Bohr magnetons (μb) for pure magnetite to 3 μb for chromian magnetite ($x = 1.38$) [*Robbins et al.*, 1971, Figure 1c]. The magnetic properties of the chromian magnetite observed in the Banded Series gabbros are comparable to low-Ti titanomagnetites, which have moderately reduced Curie temperatures and magnetic moments in comparison to pure magnetite.

The four types of iron oxide occurrences in the Stillwater gabbros include needle-shaped magnetite in plagioclase; anhedral chromian magnetite in pyroxene alteration zones and along fractures in plagioclases and pyroxenes; and cubic

Table 5. Correlation Between Magnetic Components and Magnetic Minerals

Sample	Magnetic Component	Magnetite Needles in Plagioclase	Chromian-Magnetite in cpx and Plagioclase
B14C	A	yes	no
B42C	B	no	yes
B6E	C	no	yes
B4E	C	no	yes
B4F	C	no	yes
B40I	A+?	yes	no
B32H	A+?	yes	no
B35A	A+?	yes	no
B67D	A+?	yes	no
STW3B	A+?	yes	no

yes means observed, no means not observed.
+? means a poorly defined component is present in addition to A.

euhedral magnetite in alkali feldspar veins cutting plagioclase. Iron oxides associated with sulfides (pyrrhotite or chalcopyrite) exist in almost every sample and because the sizes of these grains are generally greater than 5 μm and thus in the pseudo-single domain or multidomain range, any correlation between the presence of such iron oxides and the different magnetizations observed is not obvious. The most notable correlations we can propose between iron oxide type and NRM characteristics are based on the fact that needle-shaped magnetites within plagioclase crystals have only been observed in samples carrying the A component and that anhedral chromian magnetites have only been observed in samples carrying the B or C magnetizations. Consequently, we suggest that the preservation of an A magnetization in the Stillwater gabbros is dependent on the retention of needle-shaped magnetite grains in cumulus plagioclase; the introduction of anhedral chromian magnetite led to the acquisition of B and C magnetizations in these rocks.

Ages of Identified Magnetizations

Direct observation of the characteristics of magnetic phases in samples with geologically stable (either primary or secondary) NRM can provide information about the geologic history of the rock and facilitate correlation of magnetizations with specific geologic processes. By definition, primary NRM in igneous rocks must be carried by magnetic minerals formed during crystallization, or early alteration, at temperatures above the maximum blocking temperature and is acquired during temperature decrease over the blocking temperature range. Secondary NRM, of CRM origin, is often carried by phases whose magnetization was stabilized principally by grain growth, not by a decrease in thermal energy, below the maximum blocking temperature in a later geologic event.

Bergh's [1968, 1970] "primary" magnetization is what we label the A magnetization and current work indicates that it is of dual polarity (J.W. Geissman, unpublished data, 1996). Rock magnetic studies show that plagioclases from samples with A magnetizations have $H_{cr} \geq 65$ mT (some $H_{cr} \geq 100$ mT) and $M_r/M_s > 0.2$, suggesting that the volumetrically dominant magnetite grains are in the pseudo-single-domain grain size range [*Day et al.*, 1977]. However, the boundaries of the domain type fields on the Day's plot are also dependent on grain shape [*Borradaile et al.*, 1993]. Magnetite needles observed in unaltered plagioclase areas have high two-dimensional aspect ratios indicating that they are single-domain grains [*Evans and McElhinny*, 1969]. The magnetite needles are generally either parallel or perpendicular to the (010) plane and inclusions grown in these two orientations have minimum interface energy [*Fleet*, 1982]. The morphology (needle-shaped single crystal), chemical composition (pure iron oxide), and distinct orientations of these magnetite grains all are consistent with a high temperature subsolidus origin [*McConnell*, 1974]. The igneous origin and single domain nature of these magnetite needles are consistent with the interpretation that component A magnetizations are carried by magnetite needles and are primary magnetizations acquired during initial cooling of the Stillwater complex at circa 2.710 Ga. Banded Series rocks whose NRM consists solely of the A magnetization have provided an internally consistent paleointensity determination for the late Archean, using the double Thellier thermal method (J.W. Geissman and C.J. Hale, manuscript in preparation, 1996).

The dominant carriers of the B and C magnetizations are anhedral Cr-magnetite grains introduced during hydrothermal alteration and local serpentinization of Banded Series rocks. These magnetizations are interpreted to be secondary and, in the absence of direct geochronologic data bearing on the age of these magnetizations, likely ages can only be proposed in the context of extant Precambrian paleomagnetic data from the Precambrian craton of North America (i.e., Laurentia). The reliability of the comparison is severely hindered, because the oldest high-quality paleomagnetic data from the Wyoming Province, outside of, possibly, the A magnetization from the Stillwater complex, are from the circa 1.42 Ga Laramie Anorthosite/Sherman Granite [*Harlan et al.*, 1994] emplaced in Archean rocks of the Wyoming Province and Paleoproterozoic rocks stabilized along its southeast margin. The paleomagnetic record for the Wyoming Province prior to circa 1.42 Ga is nonexistent. The general agreement between the circa 1.42 Ga data from the Laramie Anorthosite/Sherman Granite and intrusions of similar age in the Superior and other provinces suggests that by this time the Wyoming Province was clearly stabilized with respect to other provinces of similar or older age. The Superior Province has yielded the majority of older Precambrian data for Laurentia. Although several workers argue that stabilization of the Wyoming relative to the Superior Province occurred earlier, in the Paleoproterozoic, there are no paleomagnetic data available to confirm this. With these limitations in mind, we briefly review possible ages for the B and C magnetizations in the Stillwater Complex gabbros.

We assume that both of these secondary magnetizations must be corrected for deformation related to Late Cretaceous to early Tertiary (Laramide) uplift of the Beartooth Mountains. Positive inclination B magnetizations, after correcting for moderate (50° to 70°) north-northeast side down tilt are of shallow positive inclination and north declination. This direction is comparable to results from the well-studied circa 2.45 Ga Matachewan dike swarm in the Superior Province [e.g., *Halls*, 1991] but too shallow in inclination compared to younger Paleoproterozoic results from the Superior Province [*Buchan et al.*, 1994]. A possible interpreted age for acquisition of the B magnetization is the early Paleoproterozoic (2.3 to 2.5 Ga) and limited geochronologic data suggest that at least some of the numerous mafic dikes that cut the Stillwater Complex were emplaced during this time period [*Page*, 1977].

The C magnetization, of southeast (or northwest) declination and shallow inclination, is corrected to a shallow negative to, more commonly, a moderate positive inclination and southeast declination (or antipode). Its paleomagnetic pole lies along either the descending (circa 1.85 Ga) or ascending (circa 1.70 Ga) arm of the Coronation loop in the Proterozoic apparent polar wander path for Laurentia [*Irving*, 1979], defining Hudsonian events affecting Archean and Proterozoic rocks. We again emphasize that none of the data defining this loop were obtained from Wyoming Province rocks, so our correlation is subject to considerable uncertainty. Nonetheless, we note that limited geochronologic data also suggest that the time period between about 1.8 and 1.6 Ga was marked by mafic dike emplacement in the Beartooth

Mountains [*Page*, 1977]. Elsewhere in southwest Montana, there is evidence for a tectonic event that led to local penetrative metamorphism and resetting of K-Ar and Rb-Sr mineral systems at circa 1.8 Ga [*Gilletti*, 1966; *O'Neill et al.*, 1988; *Erslev and Sutter*, 1990; *Harlan et al.*, 1996]. We favor the reasonable interpretation that acquisition of the C magnetization was in response to mafic dike emplacement and attending fluid mobilization.

Correlation Between Magnetization Events Affecting the Stillwater Complex

We propose distinct associations between (1) an interpreted primary magnetization in the Stillwater complex and Ti-free needle-shaped magnetite in unaltered plagioclase areas and (2) secondary magnetizations and anhedral Cr-magnetite as well as the absence of abundant Ti-free magnetite needles in cumulus plagioclase. The associations prompt important considerations concerning the retention of primary magnetizations in Banded Series rocks and the destruction of early formed magnetic phases in rocks with introduced secondary magnetic phases. In this context, any predictive measures of how to discriminate between mafic intrusive rocks carrying exclusively primary magnetizations and those that have been remagnetized in one form or another would be useful for paleomagnetic studies of such rocks.

The source for the chromium in secondary, anhedral magnetite is most likely the peridotite member of the Ultramafic Series, where essentially all of the cumulus chromite occurs [*Jackson*, 1961, 1963]. *Beenson and Jackson* [1969] noted that chromite in the peridotite member has been pervasively altered, with a major alteration product low-chromian magnetite $\{Fe^{2+}_{0.99}Mg_{0.1}\}\{Fe^{3+}_{1.75}Cr_{0.21}Al_{0.04}\}O_4$ identical in composition to the Cr-magnetite observed in remagnetized Banded Series mafic rocks (Table 3). Iron and chromium were likely transported by hydrothermal fluids from the Ultramafic Series rocks during serpentinization. Regardless of the source of the chromium in secondary magnetite in Banded Series rocks, our observations show an abundance of fractures along which fluids penetrated the remagnetized rocks containing secondary oxides (Figures 5e, 5f, and 7g).

Concerning the presence of magnetite needles, recognized in unaltered parts of cumulus plagioclase, we speculate that their absence requires a process by which they were destroyed in samples now characterized by secondary magnetizations. This argument, again, assumes that magnetite needles in cumulus plagioclase were acquired early in the crystallization history of Banded Series rocks. The timing of magnetite growth in plagioclases in cumulate and other mafic rocks, in relationship to remanence properties, has been the subject of some investigation [e.g., *Halls and Palmer*, 1990; *Zhang and Halls*, 1995]. In mafic dikes of the Kapuskasing area, north central Ontario, *Halls and Palmer* [1990] noted a correlation between clouding of plagioclase and depth of emplacement. Although the likely phases responsible for darkening of plagioclase were never investigated at high resolution, *Halls and Palmer* [1990] attributed the clouding to "minute inclusions of spinel, magnetite, or other minerals" and argued that the degree of clouding was related to the depth of emplacement of the dikes and attending alteration at deep, upper amphibolite to granulite grade, crustal levels. The Stillwater Complex was emplaced at a relatively shallow crustal depth. A 2 to 3 kbar total pressure estimate for pelitic hornfels rocks at the base of the complex, as summarized by *Geissman and Mogk* [1986] is consistent with magmatic fractionation patterns suggesting a shallow level of emplacement [*McCallum et al.*, 1980]. Banded Series rocks were probably no deeper than 10 km at the time of crystallization, and thus there is no compelling reason to correlate growth of fine magnetite needles in cumulus plagioclase with subsequent alteration at moderate to deep crustal levels. The relatively minor fluid interaction in Banded Series rocks led to alteration of presumably early formed oxides to chlorite and zoisite (Figures 7a and 7b). Destruction of early formed magnetite needles was a relatively subtle process in terms of fluid/rock interactions, because cumulus plagioclase grains in remagnetized rocks are largely unaltered and there does not appear to be any correlation between the presence of secondary magnetizations and significant modification to the oxygen isotope systematics of cumulus plagioclase grains. Parts of cumulus plagioclase grains located in areas with few fractures escaped hydrothermal alteration and thus preserve the primary magnetization carried by magnetite needles. However, we cannot rule out the possibility that primary magnetite needles were absent when the plagioclase crystallized in some rocks.

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